

TYLSTRUP, Denmark

WMO: 060310

Lat: 57.185N

Lon: 9.953E

Elev: 13

StdP: 101.17

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.7	-6.9	-11.8	1.4	-8.6	-9.4	1.7	-5.2	19.4	2.6	17.4	1.5	1.5	10	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	25.5	18.1	23.7	17.5	21.9	16.5	19.3	23.4	18.4	22.0	17.5	20.8	3.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.9	12.9	20.7	17.0	12.1	19.8	16.1	11.5	19.0	54.9	23.5	52.1	22.1	49.4	20.9	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	11.5	9.5	-12.6	27.6	3.6	1.9	-15.2	29.0	-17.3	30.1	-19.4	31.2	-22.0	32.6		
(5)			-12.7	20.7	3.4	0.9	-15.2	21.4	-17.2	21.9	-19.2	22.4	-21.7	23.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	0.8	0.8	3.1	7.2	11.0	13.9	16.8	16.1	13.2	9.1	5.4	2.6		
	DBStd	6.45	3.88	3.80	3.66	3.06	3.29	2.46	2.46	2.03	2.39	2.94	3.29	4.12		
	HDD10.0	1302	287	258	213	94	26	1	0	0	3	52	141	228		
	HDD18.3	3656	545	491	472	334	227	134	63	74	154	286	388	487		
	CDD10.0	712	0	0	1	10	58	119	211	189	99	24	2	0		
	CDD18.3	23	0	0	0	0	1	2	15	5	1	0	0	0		
	CDH23.3	191	0	0	0	0	8	28	131	21	2	0	0	0		
	CDH26.7	21	0	0	0	0	0	2	18	1	0	0	0	0		
(14) Wind	WSAvg	3.9	4.7	4.5	4.4	3.9	3.7	3.7	3.2	3.4	3.5	3.7	4.2	4.5		
(15) Precipitation	PrecAvg	622	46	31	35	35	45	59	63	68	66	66	60	51		
	PrecMax	875	102	88	87	81	120	124	200	202	177	153	120	110		
	PrecMin	431	4	2	1	5	5	13	6	11	21	15	14	9		
	PrecStd	107	23	19	19	18	24	32	40	39	31	32	25	25		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	8.8	13.6	19.0	24.6	26.2	28.7	25.6	22.6	17.3	13.7	10.8		
		MCWB	8.2	7.3	8.8	12.3	16.5	17.2	18.6	18.7	17.3	14.6	12.5	9.8		
	2%	DB	7.9	7.5	10.8	17.2	22.0	23.7	26.7	23.6	20.4	15.6	12.0	9.5		
		MCWB	6.9	6.3	7.7	11.1	14.8	16.7	18.6	17.9	16.2	13.7	11.1	8.7		
	5%	DB	7.0	6.6	8.9	15.0	19.8	21.5	24.7	21.9	18.7	14.5	10.9	8.6		
		MCWB	6.0	5.6	6.7	10.2	14.5	15.9	18.3	16.8	15.4	13.0	9.9	7.9		
	10%	DB	6.1	5.7	7.6	12.7	17.3	19.3	22.5	20.4	17.4	13.4	9.9	7.8		
		MCWB	5.2	4.8	5.9	9.1	13.0	14.9	17.3	16.4	14.5	12.1	9.2	6.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.4	7.8	9.6	13.7	18.1	19.3	20.6	20.1	18.2	15.4	12.6	10.0		
		MCDB	8.9	8.5	12.4	17.6	22.7	23.6	25.6	23.5	20.9	16.7	13.4	10.7		
	2%	WB	7.1	6.6	8.2	12.0	16.3	17.9	19.7	18.8	16.9	14.3	11.1	8.9		
		MCDB	7.6	7.3	10.2	16.0	20.0	22.1	24.4	21.8	19.4	15.2	11.8	9.3		
	5%	WB	6.1	5.8	7.1	10.7	14.7	16.6	18.7	17.8	16.0	13.2	10.1	8.0		
		MCDB	6.7	6.4	8.3	14.2	18.6	20.4	22.9	20.5	18.0	14.1	10.7	8.5		
	10%	WB	5.1	5.0	6.2	9.4	13.4	15.4	17.9	17.0	15.2	12.3	9.2	7.1		
		MCDB	5.8	5.7	7.3	12.1	16.7	18.3	21.5	19.4	16.8	13.2	9.9	7.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	4.2	5.7	8.0	8.7	8.3	8.6	7.8	7.0	5.9	4.3	4.1		
		MCDBR	4.3	4.5	7.2	11.6	12.5	12.7	12.6	10.3	8.8	6.2	4.2	4.1		
		MCWBR	4.4	3.8	4.8	6.7	7.0	6.9	6.0	5.2	5.4	4.7	3.9	4.2		
		MCDBR	4.5	4.4	6.3	10.5	10.8	10.9	8.5	7.6	5.6	4.2	4.2	4.2		
	5% WB	MCWBR	4.5	3.9	4.5	6.4	6.6	6.7	5.8	5.1	5.0	4.5	4.0	4.2		
(40) Clear-Sky Solar Irradiance	taub		0.307	0.332	0.350	0.387	0.379	0.369	0.389	0.390	0.364	0.346	0.323	0.292		
	taud		2.228	2.290	2.327	2.279	2.357	2.406	2.372	2.376	2.438	2.405	2.290	2.198		
	Ebn at Noon		551	708	802	824	857	871	842	814	783	688	530	455		
	Edh at Noon		52	75	93	114	112	108	110	102	83	67	49	41		
(44) All-Sky Solar Radiation	RadAvg		0.38	0.94	2.36	4.03	5.38	5.92	5.45	4.26	2.87	1.36	0.50	0.27		
	RadStd		0.08	0.17	0.21	0.46	0.47	0.45	0.52	0.29	0.28	0.17	0.09	0.03		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		+0.35	N/A	N/A	N/A	N/A	N/A	-117	-154	N/A	N/A

Nomenclature: See separate page